

Lecture 1 CPS

Descriptive Regression with a Binary Covariate

Outline

- ▶ Univariate Descriptive Statistics
- ▶ Descriptive Regression with a Binary Covariate

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- ▶ The data we have might be a sample (e.g., an exit poll) or it might be the whole population (e.g., administrative vote counts), sometimes known as a census.
- ▶ Having the whole population is becoming more common in the era of Big Data.
- ▶ Note: if you are interested in the causal effects of a treatment (a drug, a policy, etc.), you can never really have the whole population (we will discuss this in a few weeks).

What makes a good summary?

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From the google definition:

1. brief
2. includes main points

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 - ▶ minimizes the information lost by reducing the data
 - ▶ I'm using "information" in a loose sense here, will formalize a bit later
 - ▶ reducing the data means taking the data, which are often represented by $n * v$ numbers (with n the number of observations and v the number of variables) and reducing to a descriptive statistic that is represented by a few numbers.

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Enough forest. Trees (examples) are coming!



"Hold on, where's the forest again?"

It is easy in a statistics class to focus on the specifics. I want you to remember why we are doing what we are doing.

Outline

Univariate Descriptive Statistics

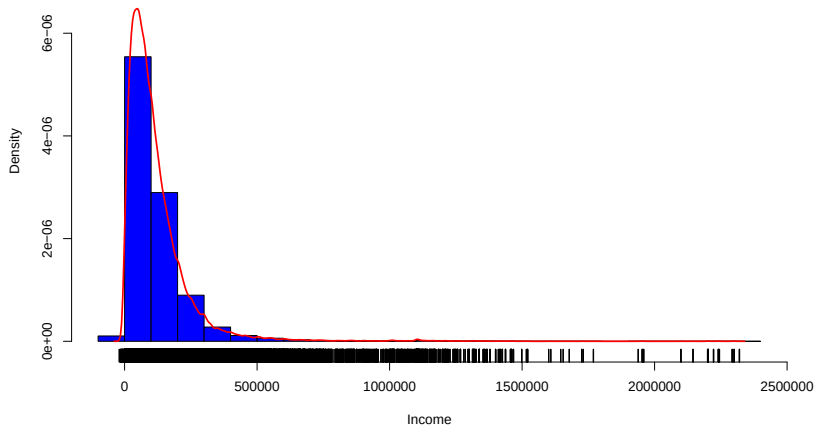
Univariate Example: CPS Wage Data

- ▶ Data on 2022 household income from the CPS

Household	income
1	44220
2	82752
152732	32300

Describing Univariate Data

Histogram of income



Description/Summarization

- ▶ Even for univariate data, we may want to focus on one or two characteristics of the population instead of describing it in its entirety.

Description/Summarization

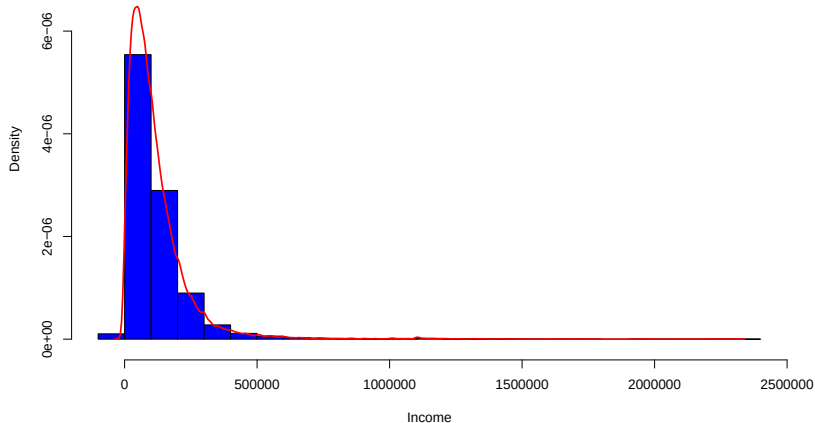
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- ▶ We typically focus on measures of center and spread.
- ▶ We will discuss the rationale for this over the coming weeks.

Univariate Activity

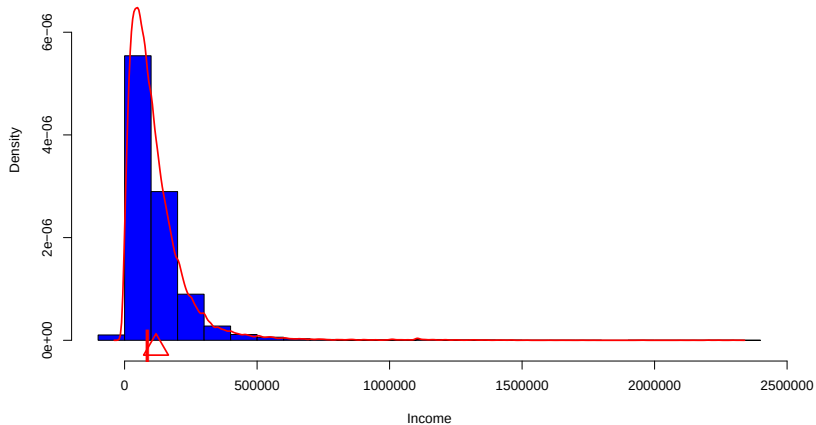
Histogram of income



Find the rough location of the mean and median.

Univariate Activity Solution

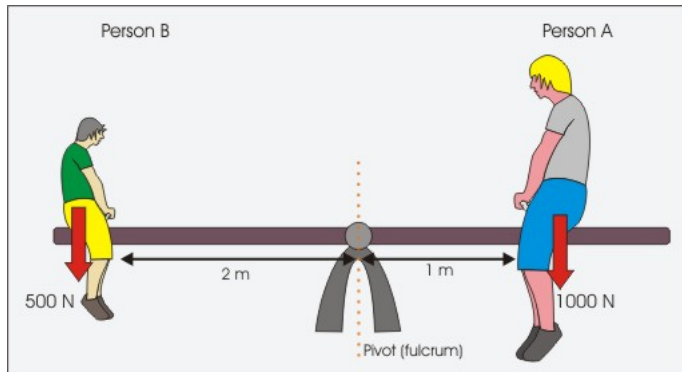
Histogram of income



Mean/Median Discussion

In pairs, discuss what the mean and median represent in the context of the household income variable. What does it mean for the mean to be larger than the median in this context?

Mean as a balance point



Choosing a Descriptive Statistic

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A Least Squares Summary

Suppose we want to pick a single number (m) for the middle of the data that summarizes all the values for y , by minimizing the sum of squared residuals (i.e., least squares).

$$SSR(\tilde{m}) = \sum_{i=1}^N (y_i - \tilde{m})^2$$

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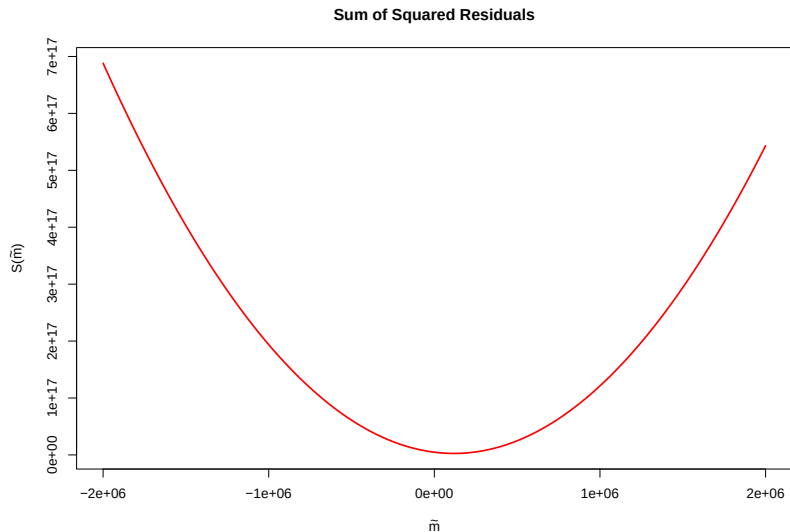
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Activity: Using calculus, derive the least squares estimator

1. Calculate the derivative of SSR with respect to $\tilde{m}$
2. Set the derivative equal to 0
3. Solve for m

The Objective Function for SSR Income

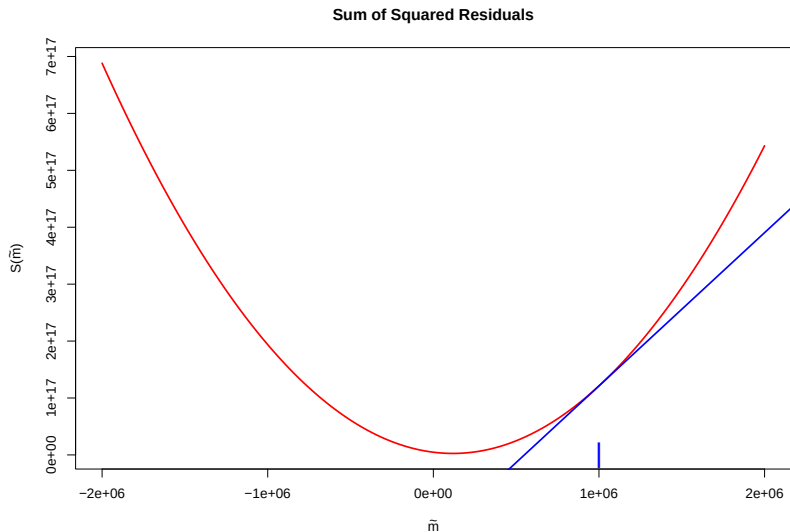


1.

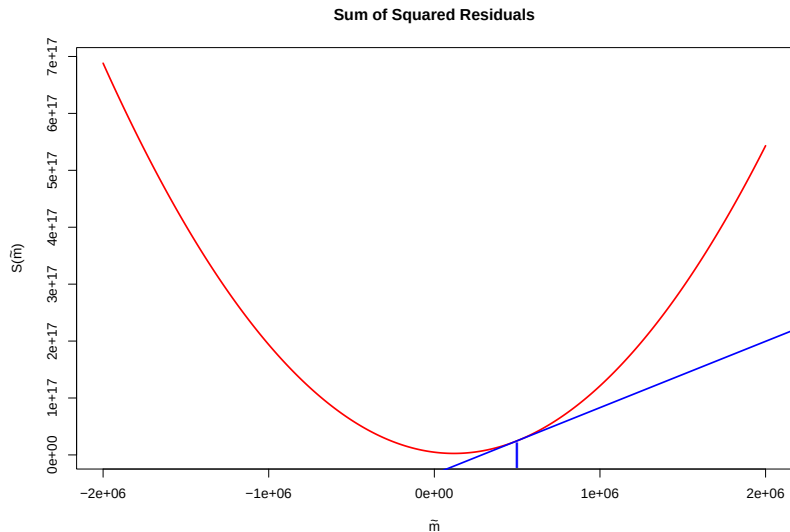
$$\begin{aligned} SSR(\tilde{m}) &= \sum_{i=1}^N (y_i - \tilde{m})^2 \\ &= \sum_{i=1}^N (y_i^2 - 2y_i\tilde{m} + \tilde{m}^2) \end{aligned}$$

$$\frac{\partial SSR(\tilde{m})}{\partial \tilde{m}} = \sum_{i=1}^N (-2y_i + 2\tilde{m})$$

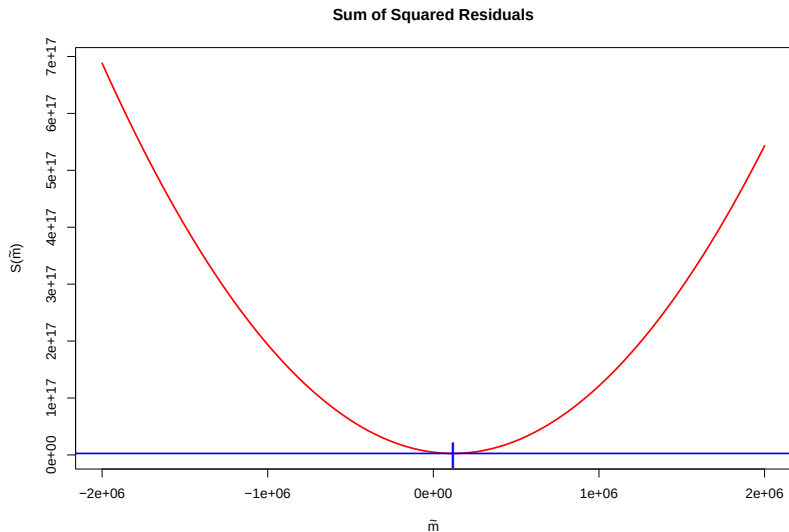
The Slope of the Tangent Line for SSR Income



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$$\begin{aligned} SSR(\tilde{m}) &= \sum_{i=1}^N (y_i - \tilde{m})^2 \\ &= \sum_{i=1}^N (y_i^2 - 2y_i\tilde{m} + \tilde{m}^2) \\ \frac{\partial S(m)}{\partial \tilde{m}} &= \sum_{i=1}^N (-2y_i + 2\tilde{m}) \end{aligned}$$

2.

$$0 = \sum_{i=1}^N (-2y_i + 2m)$$

3.

$$m = \frac{1}{N} \sum_{i=1}^N y_i$$

Therefore, the mean is a least squares summary.

An Least Absolute Value Summary

Suppose we want to pick a single number (m) for the middle of the data that summarizes all the values for y , by minimizing the sum of absolute deviations.

$$SAV(\tilde{m}) = \sum_{i=1}^N |y_i - \tilde{m}|$$

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- ▶ Why is this harder to solve?
- ▶ Any guesses?

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- ▶ The properties of means are relatively easy to describe.
- ▶ Randomized experiments only identify mean causal effects (as do regression and matching when the appropriate assumptions hold).

Variance and Standard Deviation

- ▶ Mean is the least squares predictor in terms of SSR.
- ▶ How small is “least” (i.e., how much spread around the mean)?
- ▶ Variance and Standard Deviation are useful transformations of SSR for the mean.

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Variance

$$\begin{aligned} S^2 &= \frac{\sum_{i=1}^N (y_i - \bar{y})^2}{N - 1} \\ &= \frac{SSR}{N - 1} \end{aligned}$$

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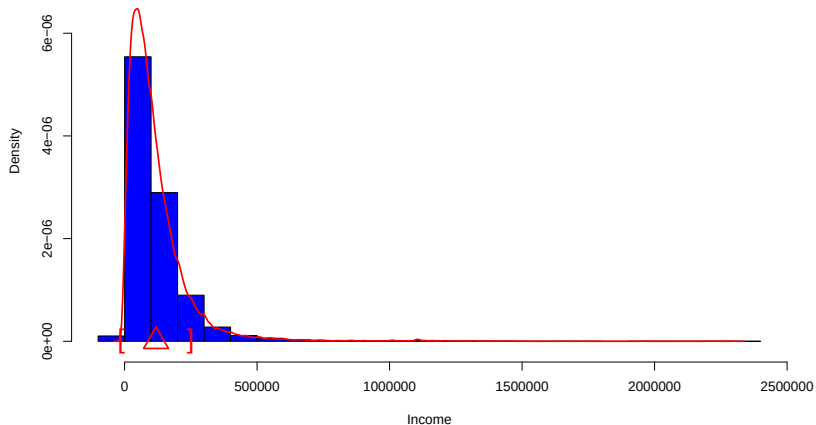
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Standard Deviation

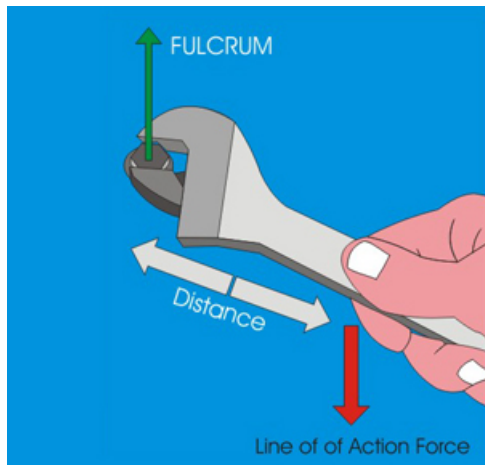
$$S = \sqrt{S^2}$$

Mean and Standard Deviation

Histogram of income



Variance as turning force



Why Variance and Standard Deviation?

- ▶ Natural measures of accuracy for means.

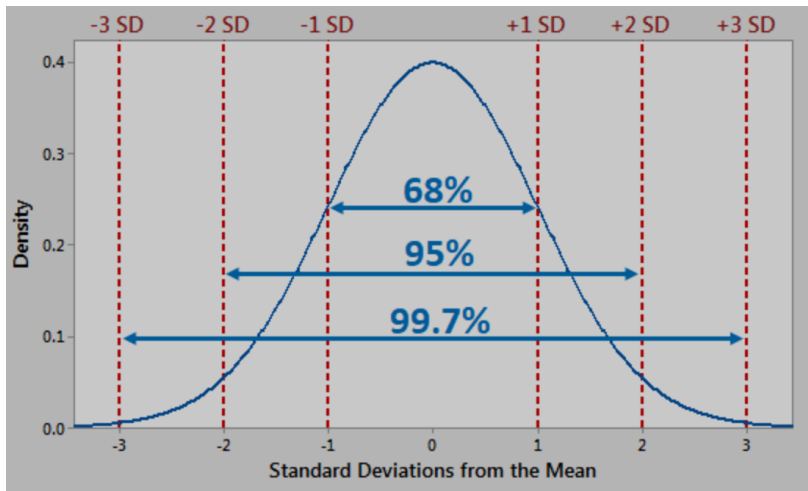
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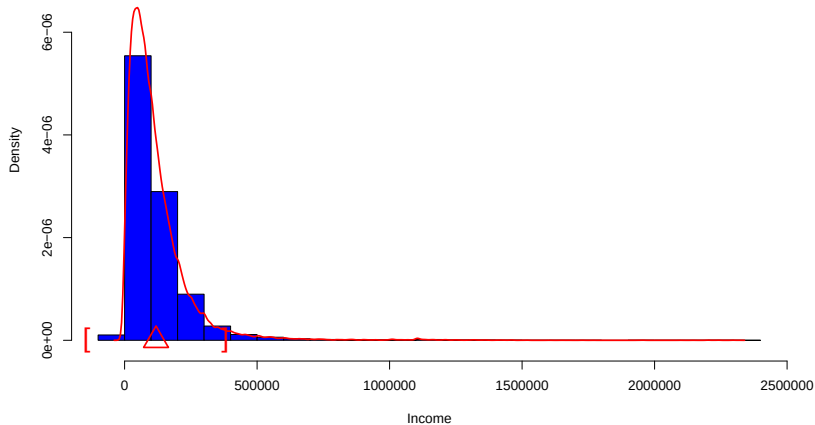
- ▶ Natural measures of accuracy for means.
- ▶ Means and standard deviations are sufficient to describe the distribution of normally distributed data.
- ▶ For non-normal data, means and standard deviations provide information via Chebyshev's inequality and other concentration inequalities.

Normally-Shaped Histograms



Chebyshev for Non-Normal Histograms

At most 25% outside of 2 SDs



Special Case: Binary Variables

Household	Education (4-year degree)
1	0
25	1
26	0
152732	0
Proportion	$38983/152732$

Mean and Variance for 4-Year Degree Variable

$$\begin{aligned}\bar{y} &= \frac{1}{N} \sum_{i=1}^N y_i \\ &= \frac{0 + \dots + 1 + 0 + \dots + 1 + 0 + \dots + 0}{152732} \\ &= \frac{38983}{152732}\end{aligned}$$

$$\begin{aligned}S^2 &= \frac{\sum_{i=1}^N (y_i - \bar{y})^2}{N - 1} \\ &= \frac{(0 - .255)^2 + \dots + (1 - .255)^2}{152732 - 1} \\ &= \left(\frac{152732}{152732 - 1} \right) [\bar{y} \cdot (1 - \bar{y})]\end{aligned}$$

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